Adult Income Classification (Model Deployment Report)

Ensembles, Imbalance Handling, Interpretability & Deployment

1. Business Goal

The objective is to predict whether an individual's annual income exceeds $50K from census demographic and employment attributes, in order to support downstream targeting decisions e.g. eligibility triage for financial products, workforce-development program targeting, or marketing segmentation. Because a positive prediction typically triggers a costly downstream action (an outreach, an offer, a review), the model favors precision on the positive class over raw accuracy, while class-imbalance handling protects recall so that high-income individuals are not systematically missed.

2. Data Used

Source: UCI/OpenML "Adult" (Census Income) dataset, 14 raw demographic and employment features (age, workclass, education, marital status, occupation, relationship, race, sex, capital gain/loss, hours-per-week, native country). After removing rows with missing values ("?") and duplicates, 45,175 records remained. The target is imbalanced: 75.2% earn ≤$50K, 24.8% earn >$50K. Six engineered features were added (capital-gain flag, log-transformed capital gain/loss, higher-education flag, education×hours interaction, marital-status flag, and binned age/hours groups) prior to a stratified 80/20 train-test split.

3. Final Model / Deployment Artifact

Three ensembles were compared (Logistic Regression baseline, Random Forest, HistGradientBoosting). HistGradientBoosting gave the best ROC-AUC (0.912) and precision (0.884) but under-served the minority class (recall 0.450) before imbalance correction. Four imbalance-handling strategies were tested inside stratified 5-fold CV using leakage-safe pipelines (class weighting, random over-sampling, random under-sampling, SMOTE) all four moved recall from ~0.45 to 0.84–0.88 at a comparable precision cost.

Final artifact: HistGradientBoostingClassifier trained with SMOTE oversampling, wrapped in a single scikit-learn/imblearn Pipeline (feature engineering → ColumnTransformer preprocessing → SMOTE → classifier), persisted as adult_income_pipeline.joblib, with a companion SHAP explainer (adult_income_shap_explainer.joblib) and inference.py providing validated single-row and batch prediction with top-3 SHAP feature attribution.

4. Key Hold-Out Metrics (SMOTE + HistGradientBoosting, threshold = 0.50)

Metric	Value
Precision (positive class)	**0.577**
Recall (positive class)	**0.847**
F1 Score	**0.686**
PR-AUC (Average Precision)	**0.792**
5-fold CV Precision / Recall	**0.583 / 0.845 (consistent with hold-out)**

SMOTE was selected over class-weighting, over-, and under-sampling because it produced the best hold-out precision–recall balance among the four (PR-AUC 0.792, F1 0.686) though the margin over random over-sampling (F1 0.680) was narrow, and SMOTE's per-fold training cost was materially higher. This trade-off is a candidate for revisiting once true inference latency budgets are known (see Section 8).

5. Chosen Threshold

The default probability threshold of 0.50 was retained for this hold-out evaluation. Precision–recall curves were generated for all candidate models but were not yet used to select a business-optimal cut-off , this is flagged as a required next step before production rollout, since the cost of a false positive (e.g. an unwarranted offer/review) vs. a false negative (a missed eligible individual) should drive the operating point, not the 0.50 default.

6. Interpretability Findings

Global (permutation importance + SHAP): the dominant predictor by a wide margin is marital status (specifically "Married-civ-spouse"), followed by the engineered education×hours interaction, age, capital gain, sex, and education-num. These six features account for the large majority of model signal; native-country and hours-per-week individually contribute little once interactions are captured.

Local (SHAP waterfall on one true positive, one false positive, one false negative): in all three cases, marital status is the largest single driver pushing predictions positive when "Married-civ-spouse" and negative otherwise. The false positive was driven by a strong education×hours interaction outweighing a non-married status; the false negative had a large positive marital-status contribution offset by negative age and education contributions. In short, the model leans heavily on a single categorical signal, which is worth stress-testing further (see Fairness).

7. Fairness Observations

Precision was compared across education levels and age buckets (using SHAP-informed protected-attribute selection would additionally include sex, which ranked among the top 8 global features but was not yet broken out separately. see next steps).

- Education: precision ranges from 0% (10th, 12th, Preschool. small, low-positive-rate groups) to 100% (1st-4th, 9th grade, very low support, 44–149 rows), a 1.00 gap. Mid/high-education groups (Bachelors–Doctorate) sit in a more reliable 0.67–0.82 band.
- Age: precision ranges from 0.346 (Young, <25) to 0.604 (Middle, 45–65) a 0.258 gap. Younger applicants are both lower-precision and lower-recall, i.e. systematically under-served.

Both gaps exceed the 0.10 disparity flag used in the notebook. Since sex appeared in the SHAP top-8 but was not evaluated as a protected group, an explicit sex/race fairness breakdown should be run before deployment, the current findings should be treated as a partial fairness audit, not a clearance.

8. Recommended Next Steps

A/B Test Plan

- Shadow-deploy the SMOTE+HGB model alongside the current process for 2–4 weeks, logging predictions without acting on them, to validate precision/recall on live traffic before any threshold or routing decision is made.
- Run a randomized A/B test on a held-out traffic slice (e.g. 10%) comparing model-assisted triage vs. the existing process, powered on the primary business metric (e.g. conversion or approval rate) rather than model precision alone.

Monitoring

- Track feature drift (PSI/KS test), prediction-rate drift, and rolling precision on newly labeled outcomes; alert on PSI > 0.20, >10% shift in predicted-positive rate, or precision dropping below 0.85 (full checklist in monitoring_checklist.csv).

Model & Process Fixes Before Production

- Fix the inference-time bug where single-row pandas Series inputs break feature engineering (np.log1p on an object-typed column) 2 of 3 unit tests currently fail for this reason.
- Re-run the imbalance comparison holding the base classifier constant across all four techniques (class-weighting was tested on Random Forest while the other three used HistGradientBoosting), and reconcile the required-columns mismatch between the notebook (drops fnlwgt) and inference.py (still requires it).
- Select an operating threshold from the precision-recall curve against a defined business cost ratio, and extend the fairness audit to sex and race before rollout.